How to Author AI Traffic

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1 Placing Static Consists

As well as authoring moving AI trains, which is detailed in the next section, static consists can be placed in the scenario. Static consists are simply items of rolling stock, or a full consist, that does not have a driver attached. This means it will have no instructions and will remain where it is placed.

Once you are in the scenario you want to edit you can place stock from the Engines and Tenders and the Rolling Stock lists.

Static consists do not have any signal occupation as this only occurs when a train has a driver, so they should not be placed on the main line or anywhere that is likely to interfere with the player or any AI trains.

Good locations for placing static consists includes yards and sidings.

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2 Setting Up AI Traffic

Moving AI (Artificial Intelligence) traffic is important in creating realistic and dynamic scenarios. Authoring the behaviour of an AI train is no different to authoring the instructions of the player train in a scenario.

Place down a locomotive from the Engines and Tenders list then complete the consist with an arrangement of wagons from the Rolling Stock menu.

Add a driver by left clicking on the Driver icon on the upper tab and then left clicking on the locomotive.

A disc with an icon of a driver will appear on the locomotive. Double click this disc to open the driver properties tab.

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Ensure that the Player Consist box remains unchecked to keep the train as an AI train. The box is unchecked by default.

At its very simplest, the contents of the driver properties tab need be all you edit to create working AI traffic. Simply specify the start time (this cannot be before the start time of the scenario), set a Service Class and set a destination.

In drive mode, when the specified start time occurs this basic AI train will follow a path to its destination where it will stop.

See Previewing AI Traffic below to ensure that the AI traffic behaves as expected without having to switch to drive mode.

More complex instructions can be given to AI trains in the same way as the player train. For passenger AI trains, you may want then to perform passenger stops.

Here are some hints to bear in mind when setting up AI traffic:

- A maximum of one train should be placed per signal occupation block.
- Stagger the start times of the AI trains so that the player will see spaced and frequent traffic.
- Set the final destinations of the AI traffic to be different from each other, or the trains will stack up.
- A good final destination for AI trains are portals. Portals are markers that be placed on the track and can then be set as a valid destination. Running an AI train into a portal as a final destination removes the AI train from the world. This has benefits for performance as well as clearing the way for other AI trains. As long as the times are staggered, the same portal can be used by multiple AI trains.
- Don't set any AI final destinations to any location that will interfere with the running of the player train.
- Don't assign AI trains the Service Class "Special" as this will give the AI train running priority over the player and could result in the player sitting at red lights.

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3 Previewing AI Traffic

Once an AI train has been placed you should check it has a valid path to its destination.

If the train has any instructions other than the driver information you can see the expected start and leave times for those instructions. If times are displayed then that AI train has a valid path to that instruction. If the destination has exclamation marks then a valid path could not be found.

If this is the case you can attempt to troubleshoot the failed pathing (see the next section), or pick a different destination.

If your AI train just has the driver instruction disc then there is no feedback whether your train has found a valid path to its destination. To test the train works you can preview it within the editor.

To preview how your AI trains work you can use the time display at the bottom of the screen:

Press the play / pause button to start and stop the passing of time in the editor. This mimics drive mode but has the added advantage of having the camera in free mode and editor functions still enabled.

Fly to your AI train and use the time display to accelerate time to when your AI train should start. The multiplier buttons accelerate time by the number on the button. If your train moves then it has a valid path.

Previewing AI train movement with the time display is also an excellent way of knowing where an AI train will be at a certain time. If you do not want to fly to the location of the AI train to watch it you can open the 2D map and scroll to the AI train there. The 2D map will show the movement of all the AI traffic.

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4 Troubleshooting Failed AI Traffic

Sometimes AI traffic can fail. Usually this is because the dispatcher cannot find a path to the destination. AI trains can also have accidents that result in an AI collision causing the scenario to end.

Here is a list of steps that will aid you in rectifying AI problems:

If the train does not path to its destination at all:

- Check the Service Class of the AI train. Freight trains for example will aim to path over track flagged as freight, whereas passenger trains avoid yard and freight track.
- Check the track directionality using the space bar visualisations. AI trains will not go against the line direction.
- Check the track for any errors that may explain why the train is not pathing where you want it to go. Track errors can include:
    - Physical breaks in the track.
    - Line type error where a section of track may be flagged as the wrong type (ie, a bit of yard type track in the middle of the mainline).
    - Junction which is unrendered and therefore does not register as a valid junction to pass over.
- Check the final destination and see if it is occupied by another AI or the player train.
- Check the destination and see if another AI or the player train is using it as a destination.
- To identify where the failure occurs, set a destination that is closer to the AI train's start position as the first instruction. If it paths to this location and the timetable is displayed then the error occurs further out. Keep changing the destination to be further out until it fails then check the track and activity of other trains in that area.
- Check the order of any instructions the AI train has. The order in which the instructions complete is from front to back. All instructions on the first disc are completed in sequence before progressing to the instructions on the next disc. Ensure that the train isn't being forced to double back on itself to complete instructions.

If the train paths to its destination along a path you are not happy with:

- You will have more control over the path the AI train takes if you add more instructions. Add a Stop At destination instruction but set the speed in the instruction to 1MPH so it acts as a waypoint and the train doesn't have to stop. Add as many waypoints as you want along the path you want the AI train to take. If there are no destination markers set up along the path you want to take you can add them via the world editor.
- Check the Service Class of the AI train. Freight trains for example will aim to path over track flagged as freight, whereas passenger trains avoid yard and freight track.

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- Check the track directionality using the space bar visualisations. AI trains will not go against the line direction.
- Check the track for any errors that may explain why the train is not pathing where you want it to go. Track errors can include:
    - Physical breaks in the track.
    - Line type error where a section of track may be flagged as the wrong type (i.e. a bit of yard type track in the middle of the mainline).
    - Junction which is unrendered and therefore does not register as a valid junction to pass over.

If there is an AI collision:

- There will be a pop up box telling you of the AI collision. Make a note of the time when the collision occurred in the scenario and replay it to just before this time then pause it. Look on the 2D map and see if your AI trains are near any static consists. If there is a static consist in the path then your AI train has probably collided with it. Because static consists have no signal occupation the AI train would not know it was there. Move the static consist to a new location or delete it
- Two AI trains can sometimes collide. Using the same method as above, look on the 2D map to see if any two trains are in close proximity at the time just before the collision. If there is, then make a note of the names of the trains and look at their driver properties. Check the Service Class and change it if it "Special" as this is a very high priority. Check the start locations of the trains; if they started in the same signal occupation block this could cause signal confusion. The performance of the trains affects their speed, so if the trains have more instructions other than the driver try adjusting the performance values so the trains don't catch each other. Finally, try adjusting the start times so the AI traffic are more staggered. If all else fails, try removing one of the offending consists.

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